Step 2: Load the Dataset

The dataset for this example is a simple CSV file containing height and weight measurements. You can load it directly from a URL or a local file using Pandas.

```
# Load dataset
url = "https://people.sc.fsu.edu/~jburkardt/data/csv/hw_200.csv"
data = pd.read_csv(url)
data.columns = ["Index", "Height", "Weight"]
```

Here, the column names are explicitly set to make the dataset more understandable.

Step 3: Display Dataset Summary

Summarizing the dataset provides insights into the data distribution, such as the mean, standard deviation, minimum, and maximum values for each column.

```
# Display dataset summary
print("Dataset Summary:")
print(data.describe())
```

This step is crucial for identifying potential issues, such as missing values or outliers, before proceeding with further analysis.

Step 4: Create a Visualization

Visualization helps in identifying patterns and relationships in the data. A scatter plot is ideal for visualizing the relationship between height and weight.

```
# Scatter plot
plt.figure(figsize=(8, 6))
plt.scatter(data["Height"], data["Weight"], color='blue', alpha=0.7, edgecolors='k') plt.title("Height vs Weight", fontsize=16) plt.xlabel("Height (inches)", fontsize=12) plt.ylabel("Weight (pounds)", fontsize=12) plt.grid(True, linestyle='--', alpha=0.5) plt.show()
```

The scatter plot includes additional customizations such as a larger figure size, transparency (`alpha`), and gridlines for better readability.

Output and Insights

The output of this script includes:

1. **Dataset Summary:** A tabular summary of key statistical metrics, including count, mean, standard deviation, min, and max values for height and weight. This helps in quickly understanding the dataset.